AAE418 Summer 2026
Syllabus

The large size of this syllabus is dictated by our Provost's office. There is no guarantee that any text below presently or always will satisfy the numerous regulations for syllabi and/or agree with content that other parts of Purdue add to the Brightspace page for this class. If unsure of what you read, talk to Professor Collicott for clarification. Project descriptions are in a separate document. Remaining in this class is a clear acknowledgement that you:

- understand that Professor Collicott may need to adjust some things to fit changes in Provost's requirements, adjust to changes that may be imposed on the Professor (e.g., remember COVID?), or simply to clarify intent.
- agree to follow safety rules and work safely, extending existing rules intelligently to adapt to new situations that can arise.
- will point out unsafe actions to teammates and classmates who may be endangering themselves or others. Safety glasses, long hair around rotating machinery, and loose clothing around rotating machinery are the most common examples of this.

Course Information

- **AAE418 Zero-gravity Flight Experiment**, Summer 2026 semester
  Sections are listed as "Lecture" and "Lab" sections or #001 and #002 04 #003 and #004 but each pair is just one three-credit class,
    - First week: class meets Monday and Tuesday in room 1109 of Neil Armstrong Hall.
    - Wednesday June 17 and after: you will meet with your team in 3098, 3191, 2106, 1109, and elsewhere as appropriate.
- 3 credit hours
- Brightspace page is available. Morning and afternoon sections are merged in Brightspace.
- Prerequisite: AAE333

Instructor(s) Contact Information

- Instructor: Professor Steven Collicott
- Office Location: 2039 Neil Armstrong Hall of Engineering
- Office hours, times and location: Most anytime we're not in class, so contact Professor or in person by email to arrange.

Course Description

Credit Hours: 3.00. Team-based design-build-test engineering experience to maximize the benefits of student participation in design, build, test of zero-gravity experiments. Gravity, orbits, and weightlessness. Low-gravity capillary fluid physics, scaling laws, and components. Experiment design for multiple short zero gravity test times. Technical proposal writing, design for manufacturability, experiment fabrication, planning, testing, and execution. Typically offered Spring, Summer, and Fall.

Unique to AAE418:

You will work as part of a small team on a real flight test experiment, drop tower, or astronaut tool. The projects are real-world and original endeavors, many funded by NASA or with expensive flights to space provided to us by rocket companies looking to build their own businesses. They are not tasks made up by your professor. Thus,

- You must attend class and there is no remote option, make-up assignments, etc., except as required by Purdue policy.
- Your work matters beyond the class! Seek the big picture.
- These payloads are real, they are not made-up plug-and-chug lab exercises nor wide-open loosely-defined paper-based design projects. They are important parts of my career and so I care very much about how we design and build them. Pay attention and talk to me.
- We have real-world flight schedules to meet.
- I manage the budgets, and they differ widely between various projects. So, if you don't get to buy what your friend on another project bought, it's not you, it's simply different budgets. The budgets range from zero to over half of a million dollars. So, expect to see major differences between spending on the projects but educational experiences in all of the projects.
- You may end up interacting with flight providers or collaborating companies. This affects my research career, so I will be demanding about you showing up on time and being prepared for these teleconferences.

Learning Resources, Technology & Texts

- Homeworks and quizzes, see Brightspace.
- Required texts: None.
- Additional readings: As assigned, and such materials are provided on Brightspace.
- Software/web resources: An unique feature will be the share drive, aae418sh, and Purdue Box, both will be discussed early in the semester.
    1. **Do not use GoogleDocs, your own hard drive, etc., for this course.**
        - Your participation grade drops when I can't access your team's files as needed for safety paperwork, verifying ordering choices, discussing sizing with rocket companies, etc. outside of class hours. Make certain that your teammates follow the rules, then you won't lose points this way.
    2. Students find that aae418sh works than Box better for CAD files unless you can install the Box desktop app on your computer, which you apparently cannot do on the computers in 2106.
    3. Solidworks is now the preferred CAD in AAE418 because it has strong support from our IT staff and many companies around the world use it. Also, Solidworks will usually read CATIA files nicely, but CATIA often does not like Solidworks files. NX is not to be used because very few students know NX.
- Tutoring support: Not applicable.
- Brightspace page: Yes, it's there. Just the "Lecture" session, not the "Lab" section as both sections form one class.

Use of AI

AI can be used in AAE418 to aid in productivity. To do this, you need to understand what AI is good at doing and use it honestly for that. You must also avoid using it for what it is not good at doing. One news article I read stated it nicely, *AI does not have reasoning powers, it merely*

repeats patterns that it has found in its training materials. That is, the I in AI is actually a lie. AI is not intelligent but it certainly does a very good job of interpolating and acting like it has intelligence.

You, on the other hand, you have intelligence. You can reason. You observe. As an engineer, you create capability that has not existed before. You can troubleshoot hardware that is sitting on the table in front of you. So, do not surrender your "superpowers" by relying mindlessly on AI in engineering and elsewhere.

The experiments in AAE418 are original experiments. They have not been done before. There is nothing for AI to have been trained on for it to know what we need to accomplish.

With this lesson about your unique powers in mind, proper uses of AI in AAE418 could be used as:

- A powerful search engine.
- Identify example circuit designs that you might adapt to create what we need.
- A method to search for other examples of sub-par performance of some component
- Locating previous work that may have similar hardware, control, etc., solutions as we are seeking
  - Note too, NASA Technical Report Server and AIAA's publications at their respective web sites are wonderful prepaid (tax dollars for NTRS and memberships for AIAA) sources of aerospace information and history for your eyes and mind to explore.

A few examples of improper use of AI would be:

- Blindly copying software.
- Writing safety documentation for our experiments
- Fabricating data, even for the simplest of benchtop testing
- Structural analysis of flight hardware
- Cheating on quizzes
- Creating short summaries of the already short reading materials in this class

So, be honest, careful, and productive.

Learning Outcomes

Specifics will vary between projects and between different times in the project from concept through flight test. But, in general, it is expected that you will dive into the project with enthusiasm, gain experience in team-based, real-world, hands-on design-build-test aerospace engineering activities on original, zero-gravity or other experiments, facilities, or technology demonstrations.

Guidance on how to succeed in the course: dive into the project, learn all you can, talk frequently to Professor Collicott about the project, listen to Professor Collicott, take notes daily, be punctual, treat teammates and their ideas with respect, act quickly on tasks in the project, report broken tooling so that it can quickly be replaced (which is much cheaper than a surprise a week before shipping a payload for launching is), be an enthusiastic self-motivated learner.

Academic Guidance in the Event a Student is Quarantined/Isolated

We're out of the covid era. But, if you become quarantined or isolated at any point in time during the semester, do email me and let me know after you have done official Purdue notification, seen your doctor, etc..

Course Schedule

We will break into teams Wednesday and get started. Background readings early in the semester.

- **Monday June 15**:
  - What we'll do, how we'll do it, etc.
  - Project descriptions questions asked and answered. You ask your questions during this time, not in the hallway or by email afterwards.
  - Background reading assignments and quizzes begin.
  - By midnight: enter your project preferences in the Brightspace survey.
- **Tuesday, June 16:** Teams assigned. More on what we'll do, how we'll do it, etc. Lab visits. More background material.
- **Friday July 3**: No Purdue classes, for July 4th weekend. 250 years of independence.
- **Monday July 6:** We have class

Schedule and assignments subject to change due to happenings in the world - remember 2020? Any changes will be emailed to you, and should but not always promptly be posted in Brightspace for redundancy.

Provost's required link to the Purdue academic calendar: Academic Calendar

Academic Integrity

The Honor Pledge Task Force, a student organization responsible for stewarding the mission of the Honor Pledge and encourages a culture of academic integrity, asks all instructors to prominently include the student-initiated Purdue Honor Pledge on their syllabus, as well as exams and key assignments. This statement is:

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, then the more information that is submitted, the greater the opportunity for the university to investigate the concern. More details are available on our course Brightspace table of contents, under University Policies.

Note: Copying, reposting, or forwarding of course materials, digital or hard copy, including to commercial sites for your profit or for assistance on homework, violates ownership of the content and will be considered academic dishonesty. See relevant paragraph in the "Grading Scale" section above. See too Appendix A below.

Nondiscrimination Statement

Throughout his 36 years at Purdue, Professor Collicott has shared in Purdue University's commitment to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. More details are apparently available on the Brightspace table of contents, under University Policies.

From the AAE Head:
The School of Aeronautics and Astronautics is also committed to a climate of inclusion; if you need to report an issue of hate or bias, you may use the link at the top right of our page here: https://engineering.purdue.edu/AAE/aboutus/Diversity/index_html .

Accessibility

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability. You are encouraged to talk to the professor and then contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247. More details are apparently available on our course Brightspace under Accessibility Information. Talk to Professor Collicott.

Mental Health Statement

In my 36 years at Purdue I've seen Purdue's on-campus Counseling and Psychological Services (CAPS) at 765-494-6995 be of genuine help to numerous students in need. I urge you to contact them in times of extreme stress, uncertainty, fear, and similar for you or for your fellow students. I certainly do care about you, yet the people at CAPS are the proven experts in helping students through many types of struggles, whereas I'm merely the expert in zero-gravity experiments.

I'm told that if you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, you could try WellTrack. Sign in and find information and tools at your fingertips, available to you at any time. I have no experiences with this to recommend it one way or the other.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 am- 5 pm.

I'm told that if you find yourself struggling to find a healthy balance between academics, social life, stress, and etc. that you can sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is completely free and can be done on BoilerConnect. If you have any questions, please contact Purdue Wellness at evans240@purdue.edu. I have no experiences with this to recommend it one way or the other.

I'm told that if you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office of the second floor of the Purdue University Student Health Center (PUSH) during business hours.

Emergency Preparation

In the event of a major campus emergency, course requirements, deadlines and grading percentages are likely going to be changed by Professor Collciott. Such changes may be necessitated by a revised Purdue semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto Brightspace or can be obtained by reading email from the Professor, the TA, or the AAE leadership. You are expected to read your @purdue.edu email on a frequent basis.

See too Appendix B below.

Reminders:

1. Keep your cell phone on and sign up to receive Purdue ALERT text messages.
2. Log into a Purdue computer connected to the network to receive any Desktop Popup Alerts.

3. Professor Collicott will email students at their Purdue email address with updates. If you prefer gmail or other, it is up to you to set up forwarding of your Purdue mail.

Appendix A - Guidelines for Academic Integrity

1. Academic dishonesty for AAE418 include but are likely not limited to:
    a. Using AI-generated work as your own,
    b. Using other peoples' work as your own,
    c. Writing a dishonest report,
    d. Purposefully hindering anyone's progress on their tasks in design, software, hardware, or similar,
    e. Collaborating on on-line quizzes or exams
    f. Coordinating peer reviews or travel-team balloting
    g. Actions which seek to give you or your friends an unfair advantage over classmates who follow the rules. That is, just because some act that you think of is not in the list above does not make it a proper act to perform in AAE418.
2. On-line quizzes are to be done:
    a. Independently. As in you working alone. Alone means no communication, collaboration, cooperation, etc. with anyone else.
    b. With reference to materials supplied to you in AAE418 this semester. Note that as there's only one or two other schools in the world that teach a class like 418 and none have the background and exam structure of AAE418, googling isn't likely to be of much help anyway in quizzes and final exam.
3. As recommended by the Provost:

    "*Incidents of academic misconduct in this course will be addressed by the course instructor and referred to the Office of Student Rights and Responsibilities (OSRR) for review at the university level. Any violation of course policies as it relates to academic integrity will result minimally in a failing or zero grade for that particular assignment, and at the instructor's discretion may result in a failing grade for the course. In addition, all incidents of academic misconduct will be forwarded to OSRR, where university penalties, including removal from the university, may be considered.*"